AIES- LAB EXPERIMENTS (16-20)

Experiment 16: Birds Fly Knowledge Base

Aim: To implement a Birds, Fly Knowledge Base using inheritance-based reasoning in Python.

CODE:

```
class Bird:
    def fly(self):
        print("Birds can fly")

class Parrot(Bird):
    pass

class Penguin(Bird):
    def fly(self):
        print("Penguins cannot fly")


p=Parrot()
q= Penguin()

print("Parrot:")
p.fly()

print("Penguin:")
q.fly()
```

OUTPUT:

```
=============== RESTAR
Parrot:
Birds can fly
Penguin:
Penguins cannot fly
>>>
```

Result: Thus, the Birds Fly Knowledge Base was implemented successfully, and inheritance-based reasoning was performed to identify flying and non-flying birds.

Experiment 17: Medical Expert System

Aim: To develop a Medical Expert System using rule-based reasoning to identify possible diseases based on patient symptoms.

CODE:

```
symptoms = input("Enter symptoms separated by space: ").lower().split()

if "fever" in symptoms and "cough" in symptoms:
    print("Possible Disease: Flu")
elif "fever" in symptoms and "headache" in symptoms:
    print("Possible Disease: Viral Infection")
elif "sneezing" in symptoms and "runny" in symptoms:
    print("Possible Disease: Common Cold")
else:
    print("Disease not identified")
```

OUTPUT:

```
========== RESTART: /Users/syedmadeehaanjum/
Enter symptoms separated by space: cough fever
Possible Disease: Flu
>
```

Result:

Thus, the Medical Expert System was implemented successfully, and possible diseases were identified using predefined rules.

Experiment 18: Fruit Color Classification

Aim: To implement Fruit Color Classification using production rules to identify fruits based on their colors.

CODE:

```
color = input("Enter fruit color: ").lower()

if color == "yellow":
    print("Fruit: Banana")
elif color == "red":
    print("Fruit: Apple")
elif color == "orange":
    print("Fruit: Orange")
elif color == "green":
    print("Fruit: Green Apple")
else:
    print("Unknown Fruit")
```

OUTPUT:

```
========== RESTART: /Us
Enter fruit color: yellow
Fruit: Banana
>
```

Result: Thus, the Fruit Color Classification was implemented successfully, and fruits were classified based on their input colors.

Experiment 19: Planet Knowledge Representation

Aim: To implement Planet Knowledge Representation using logical inference to retrieve information about planets in the solar system.

CODE:

```
planets = {
    "Mercury": "Closest to the Sun",
    "Venus": "Hottest planet",
    "Earth": "Planet with life",
    "Mars": "Red Planet",
    "Jupiter": "Largest planet",
    "Saturn": "Planet with prominent rings",
    "Uranus": "Rotates on its side",
    "Neptune": "Farthest planet from the Sun"
}

name = input("Enter planet name: ").capitalize()

if name in planets:
    print(name, ":", planets[name])
else:
    print("Planet not found")
```

OUTPUT:

```
========== RESTART: ,
Enter planet name: venus
Venus : Hottest planet
>
```

Result: Thus, the Planet Knowledge Base was implemented successfully, and planetary facts were retrieved using logical reasoning.

Experiment 20: Person DOB Knowledge Base

Aim: To implement a Person Date of Birth Knowledge Base to retrieve personal details using stored date of birth records.

CODE:

```
persons = {
    "Ali": "2005-05-10",
    "Sara": "2004-08-15",
    "Ahmed": "2006-02-20",
    "Ayesha": "2005-12-25"
}

name = input("Enter person name: ").capitalize()

if name in persons:
    print("Date of Birth:", persons[name])
else:
    print("Person not found")
```

OUTPUT:

```
========== RESTART: /Us
Enter person name: Sara
Date of Birth: 2004-08-15
>>>
```

Result: Thus, the Person DOB Knowledge Base was implemented successfully, and personal details were retrieved using logical reasoning.